$$\lim_{x\to 0} h(x) = \lim_{x\to 0}(x^2) = 0\,.$$

By Sandwich theorem,

$$\lim_{x\to 0} x^2 \sin\left(\frac{1}{x}\right) = 0.$$

We could have wrongly concluded had we resorted to applying the limit theorems, namely

$$\lim_{x\to 0} x^2\left(\sin\frac{1}{x}\right) = \lim_{x\to 0}(x^2)\lim_{x\to 0}\left(\sin\frac{1}{x}\right)$$

Now, $\lim_{x\to 0}\sin\frac{1}{x}$ does not exist, $\lim_{x\to 0} x^2 = 0$ and hence $\lim_{x\to 0} x^2\sin\left(\frac{1}{x}\right)$ leading us into trouble.

Note that, if $a \le f(x) \le a$, then $\lim_{x\to x_0} f(x) = a.$

Example 9.30

Prove that $\lim_{x\to 0}\sin x = 0$.

Solution

Since $-x \le \sin x \le x$ for all $x \ge 0$

$$\lim_{x\to 0}(-x) = 0 \text{ and}$$

$$\lim_{x\to 0}(x) = 0.$$

By Sandwich theorem

$$\lim_{x\to 0}\sin x = 0.$$

Example 9.31

Show that $\lim_{x\to 0^+} x\left[\left\lfloor\frac{1}{x}\right\rfloor + \left\lfloor\frac{2}{x}\right\rfloor + \cdots + \left\lfloor\frac{15}{x}\right\rfloor\right] = 120\,.$

Solution

$$\frac{1}{x} - 1 \le \left\lfloor\frac{1}{x}\right\rfloor \le \frac{1}{x} + 1$$

$$\frac{2}{x} - 1 \le \left\lfloor\frac{2}{x}\right\rfloor \le \frac{2}{x} + 1$$

$$\vdots$$

$$\frac{15}{x} - 1 \le \left\lfloor\frac{15}{x}\right\rfloor \le \frac{15}{x} + 1$$

Summing, we get,

$$\frac{120}{x} - 15 \le \left\lfloor\frac{1}{x}\right\rfloor + \left\lfloor\frac{2}{x}\right\rfloor + \cdots + \left\lfloor\frac{15}{x}\right\rfloor \le \frac{120}{x} + 15$$

$$120 - 15x \le x\left[\left\lfloor\frac{1}{x}\right\rfloor + \left\lfloor\frac{2}{x}\right\rfloor + \cdots + \left\lfloor\frac{15}{x}\right\rfloor\right] \le 120 + 15x$$

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